



WELCOME





Introduction to Python Programming

Recap

How to Learn a New Programming Language in Record Time

- ✓ How to Master a New Programming Language
 - Three Key Areas to Focus
- ✓ Programming Languages
 - Types of Programming Languages
- ✓ Choosing a new Programming language
 - Python



Agenda for Today's Session

Introduction to Python Programming

First Python Program

Arithmetic Operations

Python vs JavaScript

First Python Program

Introduction to Python Programming

First Python Program

Let's Write Our First Ever Python Program!

"Hello World" Program



Introduction to Python Programming

First Python Program

Before we begin with **Python**, let's first take a look at how we have written the “**Hello World**” program in **JavaScript**



JavaScript Program



JavaScript

```
console.log("Hello World");
```

Output

```
Hello world!
```


Python Program



Python

```
print("Hello World")
```

Output

```
Hello world!
```

Python vs JavaScript

Aspect	JavaScript	Python
Output Function	<code>console.log()</code>	<code>print()</code>
Semicolon	Optional	Optional
Parentheses	Required	Required
Quotes	Single or double	Single or double

Possible Mistakes in Python

Code	Possible Mistakes
<code>prnt("Hello World")</code>	Spelling Mistake in print
<code>Print("Hello World")</code>	Uppercase "P" in Print
<code>print(Hello World)</code>	Missing quotes
<code>print("Hello World"</code>	Missing Parentheses

Possible Mistakes in Python

Code	Possible Mistakes
<code>print "Hello World"</code>	Missing Parentheses
<code>print("Hello World"</code>	Missing Parentheses
<code>console.log("Hello World")</code>	Incorrect command (using Java script command in python)

Use of Semicolons in Python



Generally, In **python** we use **Semicolon** when we require to write **multiple statements** in **one line**

Use of Semicolons in Python

```
a = 5; b = 10; print(a + b);
```

Python

```
a = 5  
b = 10  
print(a+b)
```

Python

15

Output



Semicolon acts as end of the statement

Use of Semicolons in Python

Python

```
print("Hello World");
```

(or)

Python

```
print("Hello World")
```

This **will not raise an error**,
hence **semicolons** are **optional**

Arithmetic Operations

Arithmetic Operations

Operation	Operator	Code	Output
Addition	+	<code>print(2 + 2)</code>	4
Subtraction	-	<code>print(5 - 3)</code>	2
Multiplication	*	<code>print(4 * 3)</code>	12
Division	/	<code>print(10 / 2)</code>	5.0
Floor Division	//	<code>print(10 // 3)</code>	3
Modulus	%	<code>print(10 % 3)</code>	1
Exponentiation	**	<code>print(2 ** 3)</code>	8

Division by zero

Python

```
print(2 / 0)
```

Output

```
ZeroDivisionError: division by zero
```



Python vs JavaScript

JavaScript vs Python

Operation	JavaScript	Output	Python	Output
Addition	<code>console.log(2 + 2)</code>	4	<code>print(2 + 2)</code>	4
Subtraction	<code>console.log(5 - 3)</code>	2	<code>print(5 - 3)</code>	2
Multiplication	<code>console.log(4 * 3)</code>	12	<code>print(4 * 3)</code>	12

JavaScript vs Python

Operation	JavaScript	Output	Python	Output
Division	<code>console.log(10 / 2)</code>	5	<code>print(10 / 2)</code>	5.0
	<code>console.log(10 / 3)</code>	3.333	<code>print(10 / 3)</code>	3.33
	<code>console.log(2 / 0)</code>	Infinity	<code>print(2 / 0)</code>	ZeroDivisionError
Floor Division	<code>console.log(Math.floor(10 / 3))</code>	3	<code>print(10 // 3)</code>	3

JavaScript vs Python

JavaScript



`Math.floor()` is used for
floor division

Python



`//` is used for floor
division

JavaScript vs Python

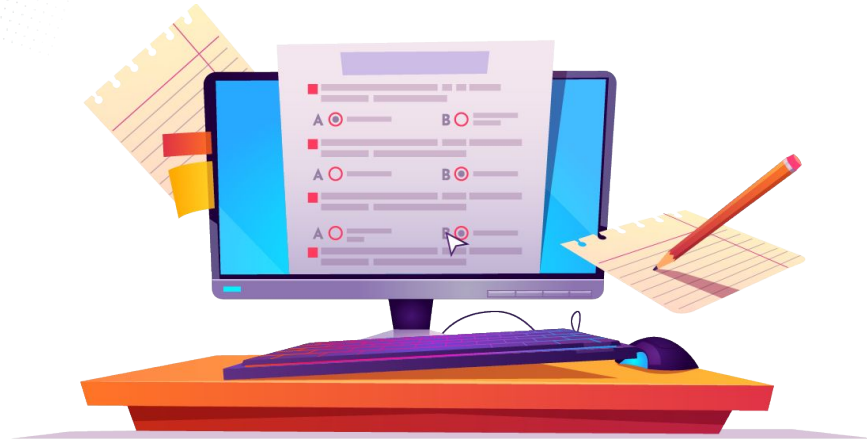
Operation	JavaScript	Output	Python	Output
Modulus	<code>console.log(10 % 3)</code>	1	<code>print(10 % 3)</code>	1
Exponentiation	<code>console.log(2 ** 3)</code>	8	<code>print(2 ** 3)</code>	8

Key Takeaways

- ✓ First Python Program
- ✓ Arithmetic Operations
- ✓ Python vs JavaScript



Practice



Feedback





Next Session

Variables and Data Types



THANK YOU





ALL THE BEST

